

A 430nW 64nV/ $\sqrt{\text{Hz}}$ Current-Reuse Telescopic Amplifier for Neural Recording Applications

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Abstract—This paper presents a low-power low-noise amplifier for neural recording applications. A single-stage current-reuse telescopic topology is proposed to achieve high DC gain and improve the noise efficiency factor (NEF) while allowing the amplifier to be scaled for high bandwidth sensing applications and/or to achieve lower thermal noise floor. The design is fabricated in a standard 0.18 μm CMOS process and occupies an active area of 0.16mm². Experimental measurements show a 430nW power consumption from a 1.2V supply, a thermal noise floor of 63.8nV/ $\sqrt{\text{Hz}}$ and a corresponding NEF of 1.5.

I. INTRODUCTION

Simultaneously recording of the cortical neurons activity enables brain-machine interfaces for people affected by debilitating neurological conditions [1]. In such applications, multiple electrodes are needed for neural pattern classification and identification. The number of channels in neural recoding systems has increased up to tens of thousands with high-density Multi Electrode Arrays (MEAs) [2], to investigate the neurons at different layers within a 3D volume [3] and enable advancements in neuroscience and pharmacology. Monitoring the electrical activity of specific brain areas using implantable neural interfaces has also demonstrated the potential to diagnose brain diseases like epileptic seizures [4], paving the way to closed-loop electric stimulation therapies of the brain.

Minimum power consumption is paramount to implement practical multiple-channel implantable neural interfaces, as low dissipation extends battery lifetime and limits heat dissipation in the implant to safe levels. The front end amplifier typically dominates the total power consumption in neural interface systems, due to the stringent equivalent input noise specifications (1-10 μV_{rms}) that are needed to cope with the signal amplitude. Designs for neural recording amplifiers with diverse specifications have shown different levels of noise efficiency factor (NEF). This figure of merit is defined in Eq. 1, where I_{tot} is the current consumption of the amplifier; V_{rms} is the input-referred in-band rms noise level, BW is the -3dB bandwidth of the signal; V_t is the thermal voltage, k and T have the usual physical meaning. NEF describes how many times the noise of a system with the same current and bandwidth is higher compared to that of an ideal bipolar transistor [5]. The lowest reported NEF to our knowledge is 1.57 [6].

$$NEF = V_{\text{rms}} \sqrt{\frac{2I_{\text{tot}}}{\pi \cdot V_t \cdot 4kT \cdot BW}} \quad (1)$$

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To obtain a good NEF, MOS input transistor(s) are biased in weak inversion, to take advantage of high g_m/I_D ratio. The theoretical limit on the NEF for a differential pair of MOS transistors is calculated to be 2.02 [1].

In this paper, a continuous-time current-reuse telescopic amplifier is proposed and implemented in a standard 0.18 μm CMOS process. This topology exploits a single stage and thus eliminates the power needed for the second stage or the folded cascode stage used in previous designs, while achieving a similar open loop gain as two-stage solutions. The single stage approach, together with current-reuse at the input devices enables a measured NEF of 1.5, which advances the state of the art. Moreover, by just tuning the bias current, the thermal noise floor can be lowered till 23nV/ $\sqrt{\text{Hz}}$ and the bandwidth of the amplifier can be scaled up to 200kHz for applications like chopping amplifiers and fast magnetic sensor readout.

The paper is organized as follows: in section II, the current-reuse telescopic topology is analyzed and compared with other amplifiers for neural recording. In section III, the detailed circuit design is discussed. Experimental results are given in section IV and conclusions are drawn in section V.

II. CIRCUIT TOPOLOGY

A. Current-Reuse Techniques

Several techniques have been proposed to improve the power efficiency in multi-channel sensor applications. The orthogonal current-reuse technique [7] stacks n input differential pairs of n folded cascode amplifiers to form a multiple-channel amplifier and thus reuses the current through the input pairs. The generated 2^n orthogonal output currents are then combined to obtain n outputs. This technique suffers from channel crosstalk and stability problems. The current sharing technique [8] reuses one non-inverting input branch in n OTAs for n channels. However the trade-off between the channel crosstalk and NEF makes it difficult to obtain a strong NEF reduction using this technique.

The current-reuse technique was first proposed for neural recording in [9], in the single-ended, open loop amplifier shown in Fig 1(a). In the core amplifier, both complementary input transistors are driven together as shown in Fig (b) (right). The equivalent g_m is then doubled assuming g_{mn} equals g_{mp} for n and p type transistors in weak inversion. As a result, the input-referred voltage noise power density is halved compared to the noise of a normal single-ended amplifier as shown in Fig. 1(b) (Left). In other words, with current-reuse,

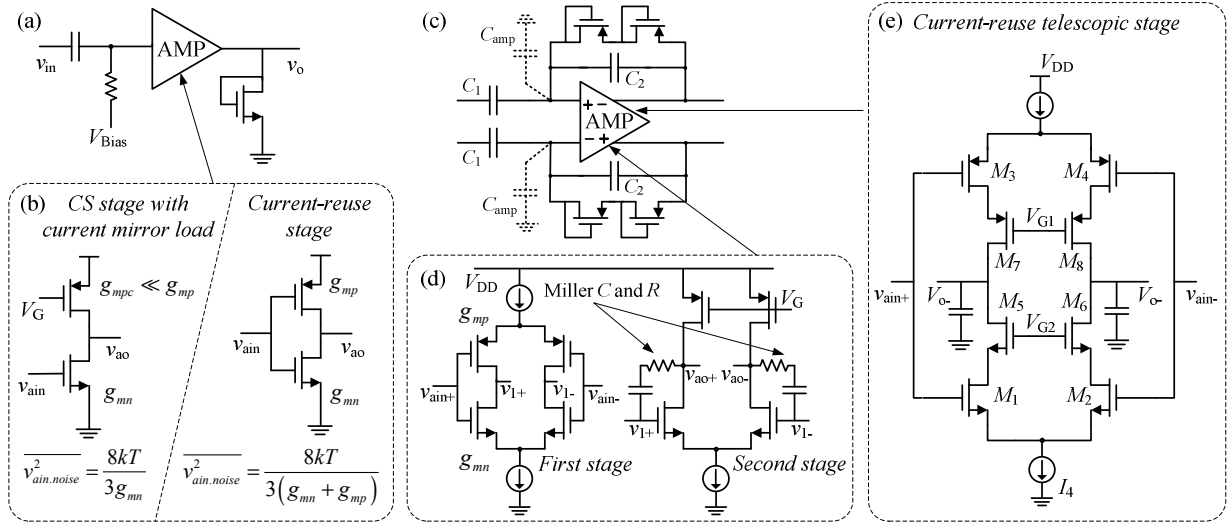


Fig. 1 (a) Single ended, open loop amplifier; (b) Current-reuse amplifier; (c) Differential feedback amplifier; (d) Current reuse, two-stage amplifier; (e) Current reuse, telescopic amplifier

the current needed to achieve a certain noise level is reduced by a factor of $\sqrt{2}$, resulting in a theoretical limit of 1.4 for the NEF of a differential pair.

A differential feedback amplifier exploiting current reuse as shown in Fig.1 (c) has been presented in [10, 11] to improve the inaccurate gain, limited linearity and PSRR observed in [9]. As shown Fig.1 (d), the core amplifier employs a differential current-reuse first stage, and a second gain stage to increase the DC gain, improving feedback gain accuracy and linearity. The noise contribution of the second stage is negligible because of the gain in the first stage. The best NEF obtained with this kind of topology is to the author's knowledge 1.57 [6]. The stability issues related to this topology will be discussed in next sub-session.

B. Current-reuse Telescopic Topology

A current-reuse telescopic topology as shown in Fig 1(e) is proposed to further improve the power efficiency of the neural amplifier and push the NEF to its theoretical limit. We use complementary input pairs (M_{1-4}) together with cascoding (M_{5-8}) in a telescopic arrangement to achieve higher gain. The voltage swing, which is usually a concern in telescopic structures, is not a problem for our application, because of the small amplitude ($<10\text{mV}$) of neural signals and the typically moderate (40dB) closed-loop gain of the first stage amplifier. The current source on the top is regulated to stabilize the common mode output level. The biasing and common-mode feedback (CMFB) circuit will be discussed in session III.

The output noise current for the first stage in Fig. 1(d) can be expressed in Eq. 2, while the output noise current of the telescopic cascade arrangement is calculated in Eq. 3, showing that the noise of M_{5-8} is negligible.

$$\overline{i_{o,noise}^2} = \frac{16kT}{3} \cdot (g_{mn} + g_{mp}) \quad (2)$$

$$\overline{i_{o,noise}^2} = \frac{16kT}{3} \cdot \left(g_{m1} + g_{m3} + \frac{g_{m5}}{1 + g_{m5}r_{o5}} + \frac{g_{m7}}{1 + g_{m7}r_{o7}} \right) \quad (3)$$

The gain of the telescopic amplifier is in the order of $(g_{m,r_o})^2$ (80dB in simulation), which is similar to that of the two-stage amplifier. The telescopic amplifier achieves thus similar gain and noise as the two-stage topology. Due to the single-stage topology the current consumption needed in the telescopic amplifier to achieve a given BW with good stability is smaller than the one of the two-stage solution. All these facts ensure a better NEF for the telescopic structure. In our $0.18\mu\text{m}$ CMOS implementation the amplifier works down to 1.0 V supply, which is compatible with other building blocks (ADC, RF links) in typical neural recording systems.

The proposed telescopic amplifier has one dominant pole at the output, showing an inherently good stability, which is largely independent on the bias current. Therefore, by simply tuning the bias current, the input-referred noise of a given amplifier can be lowered for applications like fetal monitoring and/or the bandwidth can be increased for applications like fast magnetic sensor readout. In multiple channel systems, the noise of the telescopic amplifier can be adjusted to suit the signal level at the specific recoding site and minimize the total power consumption [12]. The proposed structure can also be used as the core of a chopped amplifier for low bandwidth applications like ECG monitoring, where the $1/f$ noise can be efficiently canceled increasing the chopping frequency f_c beyond the noise corner ($\text{BW} \geq 10 \cdot f_c$ is typically required for sufficient settling).

In comparison, the two-stage amplifier shown in Fig.1(d) is a two-pole system. Miller compensation typically needs to be performed to stabilize the loop. Since the dominant pole is in the first stage, the second stage must consume current to keep its pole at high frequency and to ensure enough phase margin. Furthermore, scaling of this amplifier in bandwidth or noise level requires tuning the currents of both stages, which can be cumbersome from the biasing point of view.

III. CIRCUIT IMPLEMENTATION

A. Amplifier Core

The complete schematic of the amplifier is shown in Fig. 2, including the core, the biasing circuit for the cascode transistors M_{5-8} and the CMFB circuit. The W/L ratios of the input pairs are designed to be $150\mu\text{m}/1\mu\text{m}$ and $300\mu\text{m}/1\mu\text{m}$ for the NMOS pair and PMOS pair respectively, to make the transistor work in weak inversion and keep the $1/f$ noise corner frequency relatively low (1 kHz). The equivalent input-referred voltage noise of the feedback amplifier for both the simple and the cascaded current-reuse topologies is shown in Eq. 4, where C_{amp} is the parasitic capacitance to ground of the core amplifier as shown in Fig. 1(c).

$$\overline{v_{in,noise}^2} = \frac{16kT}{3 \cdot (g_{m1} + g_{m3})} \cdot \left(\frac{C_1 + C_2 + C_{amp}}{C_1} \right)^2 \quad (4)$$

The size of the input transistors is small enough to keep the parasitic C_{amp} to ground small and avoid an increase in $v_{in,noise}$ according to Eq. 4.

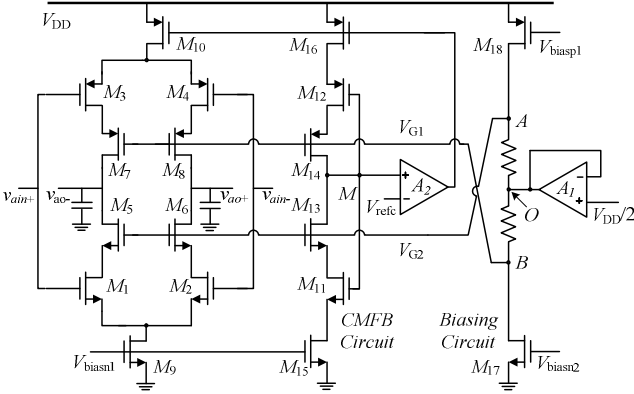


Fig. 2 Schematic of the proposed current-reuse telescopic amplifier

B. Biasing and Common Mode Feedback Circuit

The biasing circuit provides the gate voltage for the cascode device (M_{5-8}). The amplifier A_1 , connected in unity gain configuration (Fig. 2) absorbs the mismatch current between M_{17} and M_{18} at node O and keeps it at middle rail. The biasing voltages are generated from node A and B as shown in Fig. 2. The common mode voltage is sensed from a fractional replica (M_{11-16}) of the main branch at node M and compared with the reference by amplifier A_2 . Since the input biasing voltage (for M_{1-4}) is the same as the one of the output nodes, as shown in Fig. 1 (c), in the replica branch the gates of $M_{11,12}$ are also connected to node M . Thereby the current sources ($M_{10,16}$) are regulated to stabilize the common mode voltage. Careful layout guarantees that the replica is an accurate fraction of the main branch to prevent offset.

IV. EXPERIMENTAL RESULTS

The proposed current-reuse telescopic amplifier is implemented in a standard $0.18\mu\text{m}$ CMOS process together with capacitive feedback, providing a gain of 26dB. The feedback network, amplifier core, biasing, CMFB circuit and

buffers to drive the off chip load are shown on the die-photo (Fig. 3). The whole neural amplifier occupies an area of 0.16 mm^2 . The design is verified by experimental measurements.

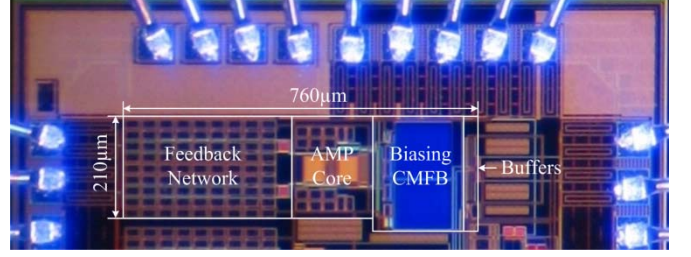


Fig. 3 Die-photo of the prototype current-reuse telescopic amplifier with capacitive feedback network

Fig. 4 shows the output voltage noise spectrum measured with Agilent HP4095a. A noise floor of $63.8\text{ nV}\sqrt{\text{Hz}}$ and a $1/f$ corner frequency of 1kHz can be derived, resulting in a V_{rms} of $8.1\mu\text{V}$ in signal band. Fig. 5 shows the measured AC transfer of the amplifier. The small-signal gain is 26dB. The -3dB bandwidth is 15kHz. The linearity is tested by a 1kHz, 10 mV_{pp} sinusoid input signal. Fig. 6 shows the output spectrum, from where a -67dB difference between the fundamental and the third order harmonics can be observed, resulting in a total harmonic distortion (THD) of 0.05%.

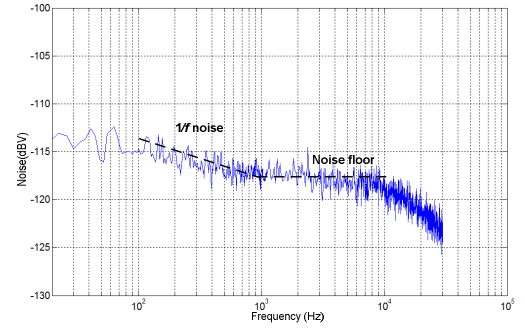


Fig. 4. Measured output voltage noise density of the amplifier

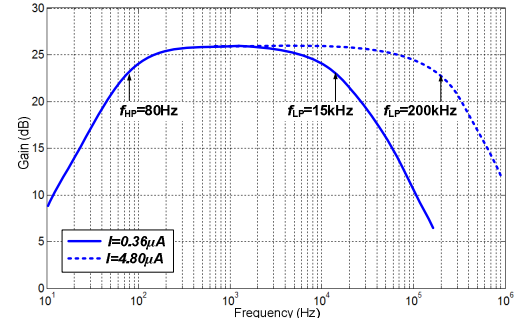


Fig. 5. Measured AC transfer of the amplifier for different bias currents

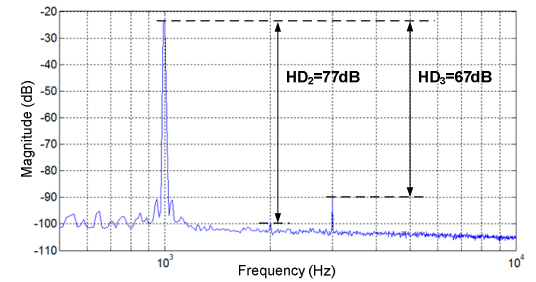


Fig. 6 Measured output spectrum the amplifier for a 10 mV_{pp} input signal

Table III. Benchmark of the proposed IA with other works

	[3]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	This work
Year	2011	2013	2010	2011	2007	2012	2012	2011	2013
Power [μ W]	0.86	0.73	7.56	7.92	0.81	0.8	12.1	2.1/16.6	0.43
Supply [V]	0.5	0.45	2.8	1.8	1.0	1.0	1.0	1.8	1.2
Gain [dB]	48	52	40.9	39.4	36.1	34/40	40	49/66	26
BW [Hz]	40 - 400	1 - 10k	45 -5.3k	10 -7.2k	0.3 -4.7k	0.2 -5.8k	0.1-11k	350-11.7k	80 -15k
Noise [μ V _{rms}]	5.32	3.2	3.06	3.5	3.6	5.71	2.2	5.4/11.2	8.1
NEF	3.09	1.57	2.67	3.35	1.8	2.59	2.9	4.4/5.9	1.52
THD [%] (mV _{inpp})	N/A	N/A	1% 7.3 mV _{pp}	1% 5.7mV _{pp}	7.1% 1mV _{pp}	N/A	1% 1mV _{pp}	N/A	0.05% 10mV _{pp}
CMRR [dB]	N/A	N/A	66	72.1	N/A	>60	80	62	>60
PSRR [dB]	N/A	N/A	75	68.3	5.5	>70	>80	72	>80
Technology [μ m]	0.13	0.18	0.5	0.18	0.5	0.18	0.13	0.18	0.18
Area [mm ²]	0.073	N/A	0.16	0.063	0.046	0.070	0.072	0.03	0.16

From measurement, tuning the bias current, the -3 dB bandwidth of the amplifier can be scaled up to 200kHz, lowering the thermal noise to 23.0nV/ $\sqrt{\text{Hz}}$ an keeping good NEF, as shown in Table II. The power consumption is 0.43 μ W and 5.76 μ W for low BW and high BW mode respectively.

Table II. Noise and Bandwidth in different current settings

Mode	Power	Noise floor	NEF	BW (-3dB)
Low BW	0.43 μ W	63.8nV/ $\sqrt{\text{Hz}}$	1.52	15kHz
High BW	5.76 μ W	23.0nV/ $\sqrt{\text{Hz}}$	2.2	200kHz

This work describes a telescopic amplifier with current-reuse technique for neural recording applications. The amplifier with a capacitive feedback is compared to previous related works in Table III. The supply voltage (1V minimum) and CMRR are well in line with other works. The gain of this work only includes the preamplifier, and can be easily increased by changing the feedback ratio (the open loop gain has been simulated to be 80dB). The measured NEF improves previous achievements.

VI. CONCLUSIONS

The design and implementation of a 430nW, 63.8nV/ $\sqrt{\text{Hz}}$ current-reuse telescopic neural recording amplifier has been presented. The amplifier uses a single-stage current-reuse telescopic topology to achieve high gain and power efficiency, while allowing the amplifier to be scaled for lower noise and/or higher bandwidth applications. The measurement results show a NEF of 1.5, which is the lowest reported compared with other works, and a THD of 0.05% at 10mV_{pp} input. Besides, the possibility to scale the bandwidth up to 200kHz with good NEF and to lower the thermal noise floor to 23nV/ $\sqrt{\text{Hz}}$ are confirmed by experimental result.

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